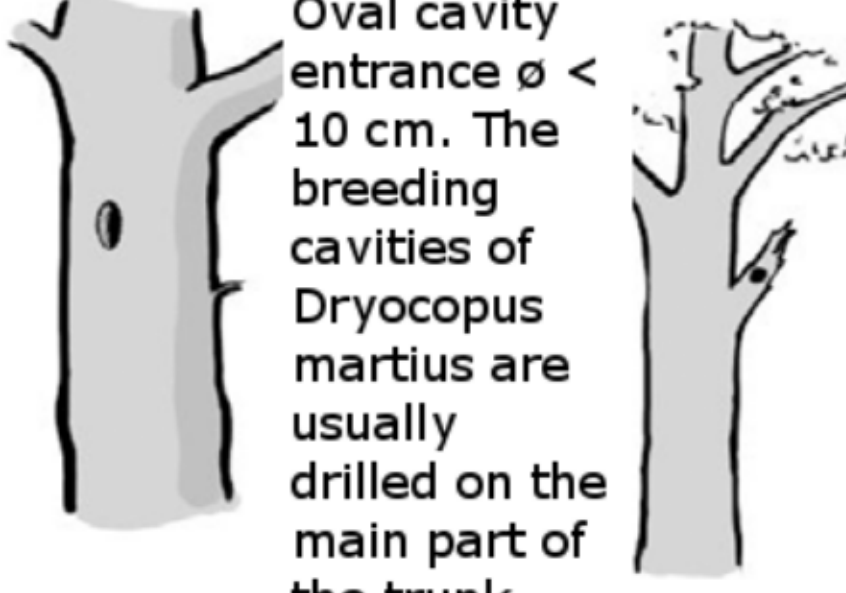
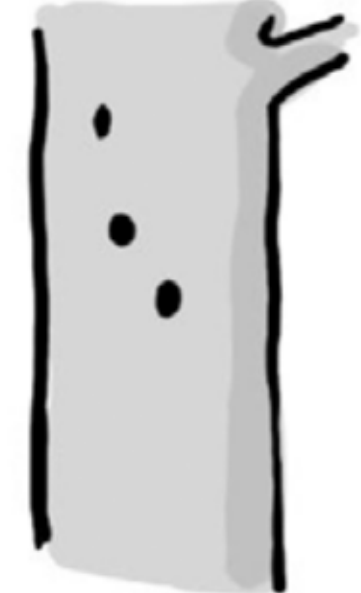
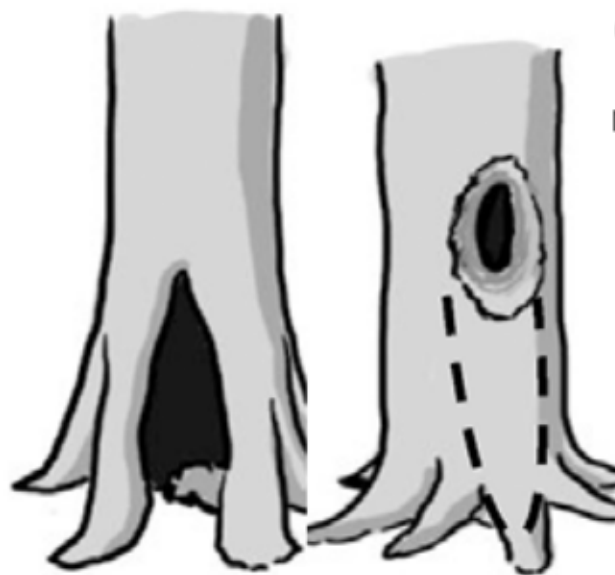
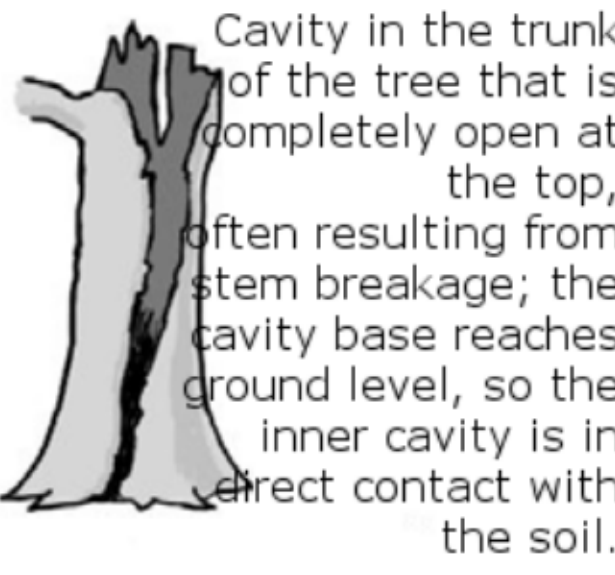
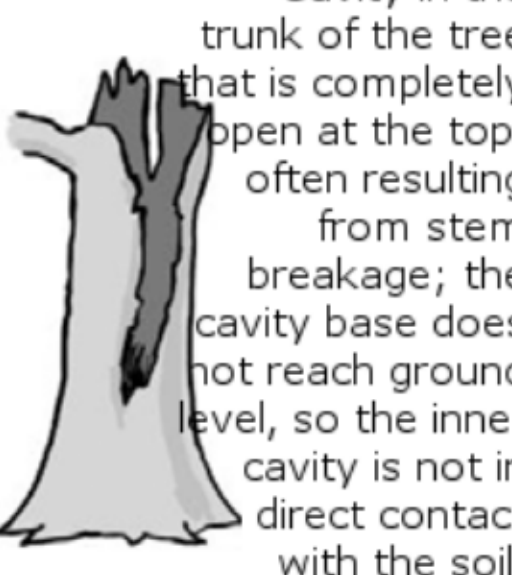
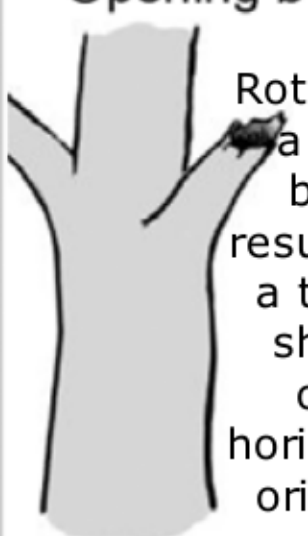
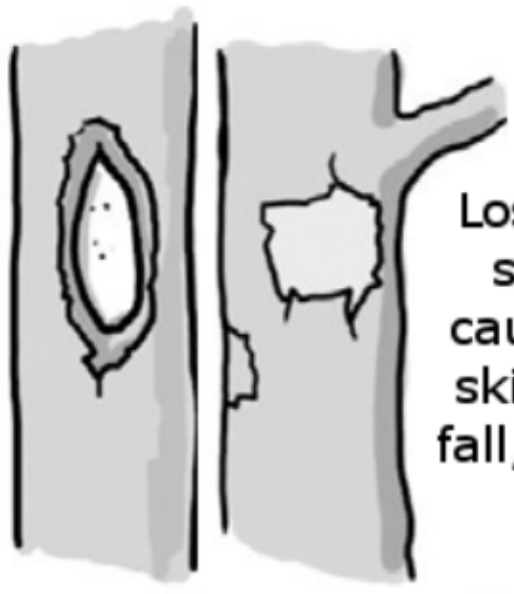
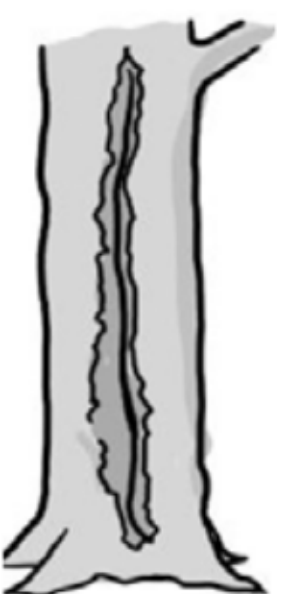
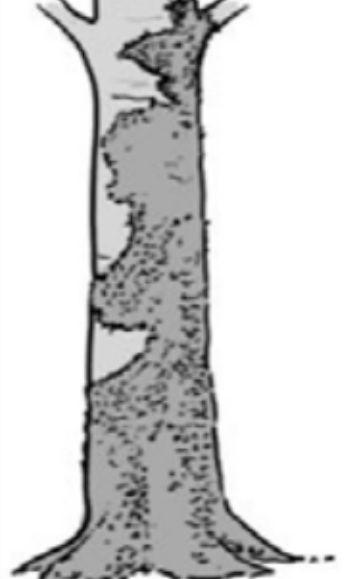
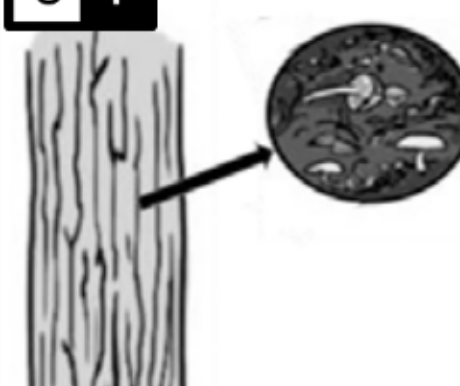


| Form                           | Group                         | Types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                          |
|--------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cavities I.s.                  | Woodpecker breeding cavities  | <div><b>1</b> <b>Small woodpecker breeding cavity</b><br/><b>123</b> Entrance <math>\varnothing &lt; 4\text{cm}</math></div> <div><p>Cavity entrance <math>\varnothing &lt; 4\text{ cm}</math>. The breeding cavity of <i>Dendrocopos minor</i> is usually drilled in a dead branch.</p></div>                                                                                                                                            | <div><b>2</b> <b>Medium-sized woodpecker breeding cavity</b><br/><b>123</b> Entrance <math>\varnothing = 4\text{--}7\text{cm}</math></div> <div><p>Round cavity entrance about <math>\varnothing = 4\text{--}7\text{ cm}</math>. The breeding cavities of the medium-sized woodpeckers are usually drilled into decaying wood (dead branch, snag, insertion of broken-off branches).</p></div> | <div><b>3</b> <b>Large woodpecker breeding cavity</b><br/><b>123</b> Entrance <math>\varnothing &gt; 10\text{cm}</math></div> <div><p>Oval cavity entrance <math>\varnothing &lt; 10\text{ cm}</math>. The breeding cavities of <i>Dryocopus martius</i> are usually drilled on the main part of the trunk.</p></div>           | <div><b>4</b> <b>Woodpecker flute</b><br/><b>123</b> Entrance <math>\varnothing &gt; 3\text{cm}</math></div> <div><p>At least three woodpecker breeding cavities in line on the trunk. Maximum distance of 2 m between two consecutive cavities.</p></div>                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                          |
|                                | Rot-holes                     | <div><b>5</b> <b>Trunk base rot-hole (closed top, ground contact)</b><br/><b>123</b> Opening <math>\varnothing &gt; 10\text{cm}</math></div> <div><p>Cavity chamber is completely protected from surrounding microclimate and rain. Top-closed trunk cavity containing more or less mould (depending on its development stage). The cavity bottom has ground contact. Note that the cavity entrance can be higher on the trunk.</p></div> | <div><b>6</b> <b>Trunk rot-hole (closed top, no ground contact)</b><br/><b>123</b> Opening <math>\varnothing &gt; 10\text{cm}</math></div> <div><p>Top-closed trunk cavity containing more or less mould (depending on its development stage). The cavity bottom has no ground contact.</p></div>                                                                                              | <div><b>7</b> <b>Semi-open trunk rot-hole</b><br/><b>123</b> Opening <math>\varnothing &gt; 30\text{cm}</math></div> <div><p>Cavity chamber is not completely protected from surrounding microclimate and rain may flow in. Note that the cavity entrance can be higher up in the trunk.</p></div>                              | <div><b>8</b> <b>Chimney trunk base rot-hole</b><br/><b>123</b> Opening <math>\varnothing &gt; 30\text{cm}</math></div> <div><p>Cavity in the trunk of the tree that is completely open at the top, often resulting from stem breakage; the cavity base reaches ground level, so the inner cavity is in direct contact with the soil.</p></div> | <div><b>9</b> <b>Chimney trunk rot-hole</b><br/><b>123</b> Opening <math>\varnothing &gt; 30\text{cm}</math></div> <div><p>Cavity in the trunk of the tree that is completely open at the top, often resulting from stem breakage; the cavity base does not reach ground level, so the inner cavity is not in direct contact with the soil.</p></div> | <div><b>10</b> <b>Hollow branch</b><br/><b>123</b> Opening <math>\varnothing &gt; 10\text{cm}</math></div> <div><p>Rot hole in a large branch, resulting in a tubular shelter, often horizontally oriented.</p></div> |
|                                | Insect galleries              | <div><b>11</b> <b>Insect galleries and bore holes</b><br/><b>01</b> Hole <math>\varnothing &gt; 2\text{cm}</math> or area <math>&gt; 300\text{cm}^2</math></div> <div><p>A bore hole network of xylophagous insects indicates a wood hole system. An insect gallery is a complex system of holes and chambers created by one or more insect species in the wood.</p></div>                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                          |
|                                | Concavities                   | <div><b>12</b> <b>Dendrotelm</b><br/><b>123</b> <math>\varnothing &gt; 15\text{cm}</math></div> <div><p>Cup-shaped concavity that, due to its form, retains water until it dries out by evaporation.</p></div>                                                                                                                                                                                                                          | <div><b>13</b> <b>Woodpecker foraging excavation</b><br/><b>123</b> Depth <math>&gt; 10\text{cm}</math>, <math>\varnothing &gt; 10\text{cm}</math></div> <div><p>Concavity resulting from the foraging activities of woodpeckers. The excavation is conical: the entrance is larger than the interior.</p></div>                                                                             | <div><b>14</b> <b>Trunk bark-lined concavity</b><br/><b>123</b> Depth <math>&gt; 10\text{cm}</math>, <math>\varnothing &gt; 10\text{cm}</math></div> <div><p>Natural bark-lined concavity on the tree trunk. No mould.</p></div>                                                                                              | <div><b>15</b> <b>Root-buttress concavity</b><br/><b>123</b> Entrance <math>\varnothing &gt; 10\text{cm}</math></div> <div><p>Natural bark-lined concavity at the base of the tree trunk formed by the tree roots and the soil. No mould (if so: see Trunk base rot hole)</p></div>                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                          |
| Tree injuries and exposed wood | Exposed sapwood only          | <div><b>16</b> <b>Bark loss</b><br/><b>123</b> Area <math>&gt; 300\text{cm}^2</math></div> <div><p>Loss of bark exposing sapwood (skinning caused e.g. by felling, skidding, natural tree fall, rock fall, rodents).</p></div> <div>counts also on snags, but only such bark loss that were established during previous living stage of tree</div>                                                                                      | <div><b>17</b> <b>Fire scar</b><br/><b>123</b> Area <math>&gt; 600\text{cm}^2</math></div> <div><p>Fire scars on the lower trunk. They usually have a triangular shape and are located at the base of the tree on the leeward side. Fire scars are associated with charcoal and sometimes resin flow on exposed sapwood or bark.</p></div>                                                   | <div><b>18</b> <b>Bark shelter</b><br/><b>123</b> Gap <math>&gt; 1\text{cm}</math>, depth <math>&gt; 10\text{cm}</math>, height <math>&gt; 10\text{cm}</math></div> <div><p>Space between peeled-off bark and sapwood forming a shelter (open at the bottom).</p></div>                                                       | <div><b>19</b> <b>Bark pocket</b><br/><b>123</b> Gap <math>&gt; 1\text{cm}</math>, width <math>&gt; 10\text{cm}</math>, height <math>&gt; 10\text{cm}</math></div> <div><p>Space between peeled-off bark and sapwood forming a pocket (open at the top) possibly containing mould.</p></div>                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                          |
|                                | Exposed sapwood and heartwood | <div><b>20</b> <b>Stem breakage</b><br/><b>01</b> <math>\varnothing &gt; 10\text{cm}</math> at break point</div> <div><p>The stem has broken off but the tree is still alive. The lower part of the deadwood is in contact with living wood with sap flow.</p></div>                                                                                                                                                                    | <div><b>21</b> <b>Limb breakage</b><br/><b>123</b> Exposed heartwood <math>&gt; 300\text{cm}^2</math></div> <div><p>Exposed heartwood through limb or fork breakage. The wound is surrounded by living wood with sap flow.</p></div>                                                                                                                                                         | <div><b>22</b> <b>Crack</b><br/><b>123</b> Length <math>&gt; 30\text{ cm}</math>, width <math>&gt; 1\text{ cm}</math>, depth <math>&gt; 10\text{ cm}</math></div> <div><p>Crack through the bark and the wood (if caused by lightning strike, see below).</p></div>                                                           | <div><b>23</b> <b>Lightning scar</b><br/><b>01</b> Length <math>&gt; 30\text{ cm}</math>, width <math>&gt; 1\text{ cm}</math>, depth <math>&gt; 10\text{ cm}</math></div> <div><p>Crack caused by lightning strike; usually spiraling around the tree with splintered wood present.</p></div>                                                 | <div><b>24</b> <b>Fork split at insertion</b><br/><b>123</b> Length <math>&gt; 30\text{ cm}</math></div> <div><p>Crack at the insertion of a fork. (If one side of the fork has broken off, see Stem breakage).</p></div>                                                                                                                           |                                                                                                                                                                                                                                                                                                          |
|                                | Crown deadwood                | <div><b>25</b> <b>Dead branches</b><br/><b>01</b> Branch <math>\varnothing &gt; 10\text{cm}</math>, or Branches <math>\varnothing &gt; 3\text{cm}</math> and <math>&gt; 10\%</math> of the crown is dead</div> <div><p>Dead branches located in the canopy, conditions remain relatively shady.</p></div>                                                                                                                               | <div><b>26</b> <b>Dead top</b><br/><b>01</b> <math>\varnothing &gt; 10\text{cm}</math> at the base of the piece of deadwood</div> <div><p>The entire top of the tree is dead; the deadwood is sun-exposed</p></div>                                                                                                                                                                          | <div><b>27</b> <b>Remaining broken limb</b><br/><b>123</b> broken end <math>\varnothing &gt; 20\text{cm}</math>, length of remaining piece <math>&gt; 0.5\text{m}</math></div> <div><p>A limb has broken off. The remaining end may be splintered. The injury does not affect the trunk (If so, see Stem breakage).</p></div> | <div><b>123</b> countable (values of 0,1,2,3 ...)</div> <div><b>01</b> uncountable (values of 0 or 1 only; =absence or presence)</div>                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                          |

| Form                                                 | Group                                           | Types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                              |
|------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Excrecences                                          | Twig tangles                                    | <div>28</div> <div>Witch broom</div> <div>Largest ø &gt;50cm</div> <div></div> <div>Dense agglomeration of twigs on branches.</div>                                                                                                                                                                                                                                                                                                                             | <div>29</div> <div>Epicormic shoots</div> <div>&gt;5 twig clusters</div> <div></div> <div>Dense agglomeration of twigs along the trunk.</div>                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                              |
|                                                      | Burrs and cankers                               | <div>30</div> <div>Burr</div> <div>Largest ø &gt;20cm</div> <div></div> <div>Proliferation of cell growth with rough bark</div>                                                                                                                                                                                                                                                                                                                                 | <div>31</div> <div>Canker</div> <div>Largest ø &gt;20cm or large part of trunk covered</div> <div></div> <div>Decayed canker. Sapwood exposed. Caused by e.g. Melampsorella caryophyllacerum, Nectria l. s.</div>                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                              |
| Fruiting bodies of saproxylic fungi and slime moulds | Perennial fungal fruiting bodies                | <div>32</div> <div>Perennial polypore</div> <div>Largest ø &gt;5cm</div> <div></div> <div>Tough fruiting bodies of perennial polypores, showing distinct annual tube layers. Main perennial genera: Fomitopsis pp, Fomes, Perreniporia pp., Oxyporus, Ganoderma pp, Phellinus, Daedalea, Haploporus, Heterobasidion, Hexagonia, Laricifomes, Daedleopsis.</div> <div></div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                              |
|                                                      | Ephemeral fungal fruiting bodies                | <div>33</div> <div>Annual polypore</div> <div>Largest ø &gt;5cm or cluster of &gt; 10 fruiting bodies</div> <div></div> <div>Fruiting bodies of annual polypores, lasting several weeks. The European annual polypores have only one layer of tubes and are usually <b>elastic and soft</b> (no woody parts).</div> <div></div>                                            | <div>34</div> <div>Pulpy agaric</div> <div>Largest ø &gt;5cm or cluster of &gt; 10 fruiting bodies</div> <div></div> <div>Large, thick and pulpy or rather fleshy fruiting body of gill- bearing fungi (order Agaricales). E.g.: Armillaria, Pleurotus, Pholiota, or large Pluteus species. The fruiting body generally remains several weeks.</div> <div></div> | <div>35</div> <div>Large Pyrenomycete</div> <div>Stroma ø &gt;3cm or stroma cluster covering &gt;100cm²</div> <div></div> <div>Tough hemispheric dark fungi resembling a lump of coal</div> <div></div> | <div>36</div> <div>Myxomycetes</div> <div>Largest ø &gt;5cm</div> <div></div> <div>Amoeboid slime mold which forms moving plasmodium. The plasmodium is gelatinous when fresh.</div> <div></div> |                                                                                                                                                                                                                                                              |
| Epiphytic and epixylic structures                    | Epiphytic and parasitic crypto- and phanerogams | <div>37</div> <div>Bryophytes</div> <div>&gt;10% of the trunk area covered</div> <div></div> <div>Trunk covered by mosses and liverworts.</div>                                                                                                                                                                                                                                                                                                               | <div>38</div> <div>Foliose and fruticose lichens</div> <div>&gt;10% of the trunk area covered</div> <div></div> <div>Trunk covered by foliose or fruticose lichens.</div>                                                                                                                                                                                                                                                                            | <div>39</div> <div>Ivy and lianas</div> <div>&gt;10% of the trunk area covered</div> <div></div> <div>Lianas and other climbing phanerogams (Hedera helix, Clematis vitalba, Lonicera periclimenum, Vitis vinifera).</div>                                                                   | <div>40</div> <div>Ferns</div> <div>&gt; 5 fronds</div> <div></div> <div>Ferns growing directly on a part of a tree (i.e. epiphyte)</div>                                                                                                                                             | <div>41</div> <div>Mistletoe</div> <div>Largest ø &gt;20cm</div> <div></div> <div>Hemiparasitic plants (Viscum spp., Arceuthobium oxycedri, Loranthus europaeus).</div> |
|                                                      | Nests                                           | <div>42</div> <div>Vertebrate nest</div> <div>ø &gt;10cm</div> <div></div> <div>Nest built by birds, dormice, mice or squirrels.</div>                                                                                                                                                                                                                                                                                                                        | <div>43</div> <div>Invertebrate nest</div> <div>Presence</div> <div></div> <div>Larval nest of invertebrates: e.g. Pine processionary moth Thaumetopoea pityocampa, wood ant Lasius fuliginosus or wild bees Apis mellifera.</div>                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                              |
|                                                      | Microsoils                                      | <div>44</div> <div>Bark microsoil</div> <div>Presence</div> <div></div> <div>Microsoil resulting from micro-pedogenesis of epiphytic mosses, lichens or algae and necrosed old, thick bark.</div>                                                                                                                                                                                                                                                             | <div>45</div> <div>Crown microsoil</div> <div>Presence</div> <div>Microsoil resulting from pedogenesis of debris and litter fallen from the crowns, often colonized by roots of the TreM bearing- tree. Main positions: flat areas on limbs, forks, sometimes stem junctions of twin</div> <div></div>                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                              |
| Exudates                                             | Exudates                                        | <div>46</div> <div>Sap run</div> <div>Cumulative length &gt;10 cm</div> <div></div> <div>Fresh significant flow of sap.</div>                                                                                                                                                                                                                                                                                                                                 | <div>47</div> <div>Heavy resinosis</div> <div>Cumulative length &gt;10 cm</div> <div></div> <div>Fresh significant flow of resin.</div>                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                              |

Larrieu, L., Paillet, Y., Winter, S., Bütler, R., Kraus, D., Krumm, F., ... & Vandekerckhove, K. (2018). Tree related microhabitats in temperate and Mediterranean European forests: A hierarchical typology for inventory standardization. Ecological Indicators, 84, 194-207.

zpracoval Radek Bače, 21.11.2018